CURVE STUDY I • QUADRATIC B-SPLINE MECHANICS | CURVATURE COMBS κ(t) & IMPLIED MIDPOINTS | KAREN CHENG EXTREMA AUDIT

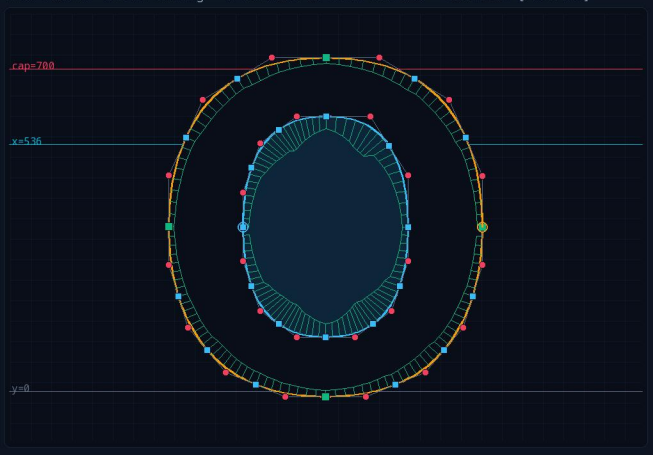
DIRECT QUADRATIC BÉZIER CURVE ATLAS

Live TrueType glyf extraction: On-Curve Extrema, Off-Curve Control Handles, Implied Midpoints & Curvature Combs

LEGEND: ■ Green = Extremum ■ Cyan = On-Curve Node ● Rose = Off-Curve Handle ◆ Gold = Implied Midpoint — Emerald Spikes = Curvature Comb κ(t)

01 / CAPITAL 'O' (PocketGull-Bold)

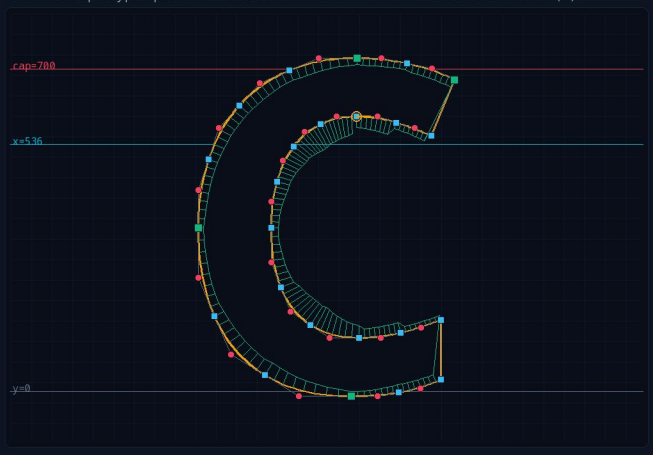
Dual-contour CW/CCW winding • 4 cardinal on-curve extrema • Overshoot [-12..712]



Contours: 2 • On-Curve ■: 25 • Off-Curve ●: 25 • Implied ◆: 0 • AdvW: 791
BBox: x=[55..735] (680 UPM) y=[-10..725] (735 UPM)

02 / LOUISE SLOAN 'C' (PocketGull-Bold)

1:5 Sloan optotype aperture • Sheared humanist terminals • Curvature comb κ(t)



Contours: 1 • On-Curve ■: 24 • Off-Curve ●: 22 • Implied ◆: 0 • AdvW: 651
BBox: x=[55..611] (556 UPM) y=[-10..724] (734 UPM)

03 / SPINE INFLECTION 'S' (PocketGull-Bold)

Continuous implied midpoint spline along diagonal spine inflection



Contours: 1 • On-Curve ■: 36 • Off-Curve ●: 34 • Implied ◆: 0 • AdvW: 551
BBox: x=[46..511] (465 UPM) y=[-10..724] (734 UPM)

04 / DOUBLE-STORY 'a' (PocketGull-Regular)

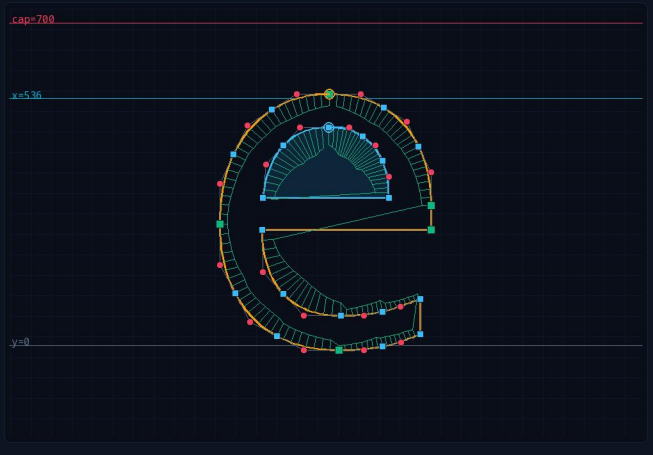
Humanist terminal hook & teardropped lower bowl counter balance



Contours: 2 • On-Curve ■: 31 • Off-Curve ●: 22 • Implied ◆: 0 • AdvW: 561
BBox: x=[46..480] (434 UPM) y=[-10..545] (555 UPM)

05 / CLINICAL APERTURE 'e' (PocketGull-Regular)

High crossbar eye & wide lower aperture preventing low-DPI blur



Contours: 2 • On-Curve ■: 24 • Off-Curve ●: 20 • Implied ◆: 0 • AdvW: 564
BBox: x=[55..513] (458 UPM) y=[-10..546] (556 UPM)

06 / LOOP & LINK 'g' (PocketGull-Regular)

3-contour topology: upper bowl, lower loop counter & humanist ear



Contours: 2 • On-Curve ■: 39 • Off-Curve ●: 31 • Implied ◆: 0 • AdvW: 615
BBox: x=[55..530] (475 UPM) y=[-240..546] (786 UPM)

CURVE STUDY II • ISMP & PILLAR 9 TOPOLOGY

COUNTER WINDING ORDER (CW vs CCW)

48 UPM TITTLE AIR GAP VERIFIED

ISMP & PHILOCARDIA CONTOUR MECHANICS

Forensic contour verification of Slashed Zero (0/), Curved l, Biserif I, Crossbar Z, and Pillar 9 Heart Tittles (ss07)

WINDING INVARIANT: — Cyan Contour = Clockwise (Outer Solid Stroke) — Amber Contour = Counter-Clockwise (Inner Counter Aperture) ◆ Gold = Implied On-Curve Midpoint

01 / SLASHED ZERO '0/' (cv08 / Numerics)

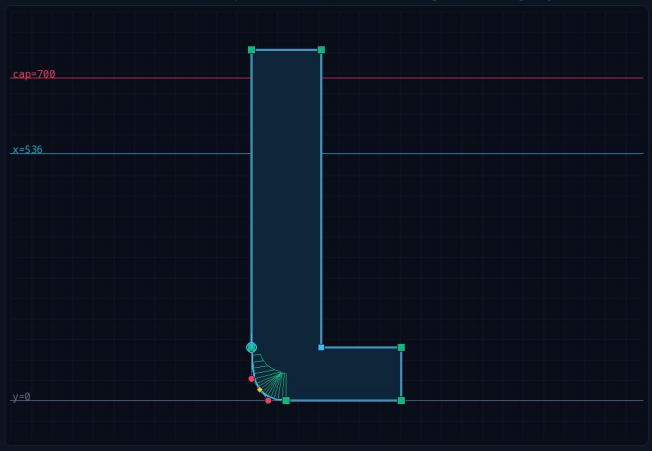
3 contours: outer oval (CW) + split upper/lower interior counters (CCW)



Contours: 3 • On-Curve ■: 22 • Off-Curve ●: 25 • Implied ◆: 5 • AdvW: 575
BBox: x=[36..535] (499 UPM) y=[-10..725] (735 UPM)

02 / CURVED FOOT LOWERCASE 'l' (cv05)

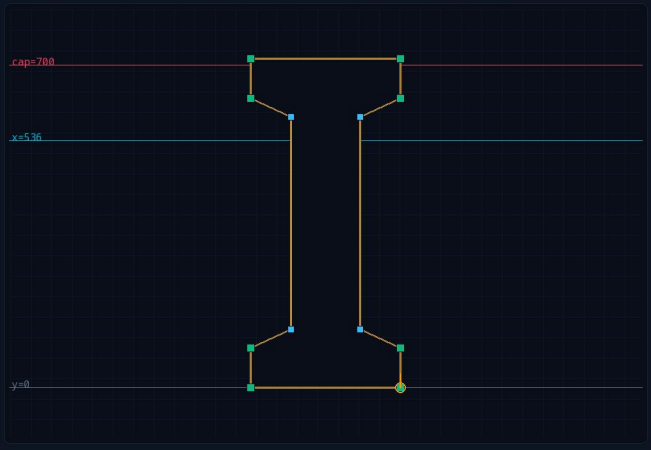
Prominent outward baseline sweep (xMax=292 UPM) eliminating 1/l/I ambiguity



Contours: 1 • On-Curve ■: 7 • Off-Curve ●: 2 • Implied ◆: 1 • AdvW: 470
BBox: x=[73..397] (324 UPM) y=[0..760] (760 UPM)

03 / BISERIF CAPITAL 'I' (ss02)

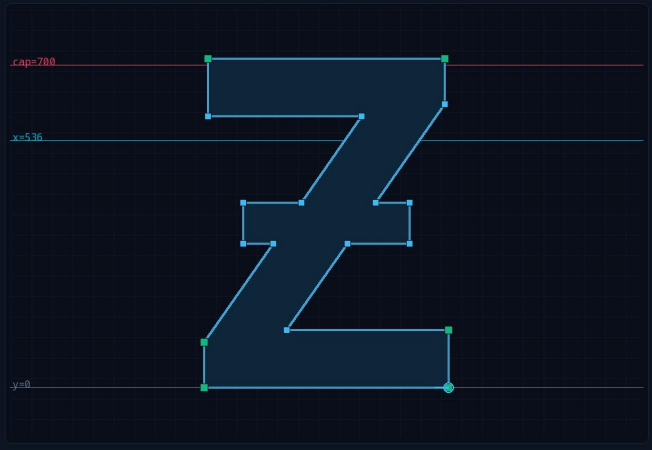
Top & baseline horizontal bilateral serifs for biomarker & IU clarity



Contours: 1 • On-Curve ■: 12 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 397
BBox: x=[32..357] (325 UPM) y=[0..714] (714 UPM)

04 / CROSSBAR CAPITAL 'Ƶ' (cv11 / Numerics)

Central horizontal crossbar stroke preventing Z-to-2 misreadings



Contours: 1 • On-Curve ■: 18 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 595
BBox: x=[24..555] (531 UPM) y=[0..714] (714 UPM)

05 / PHILOCARDIA 'i.heart' (ss07)

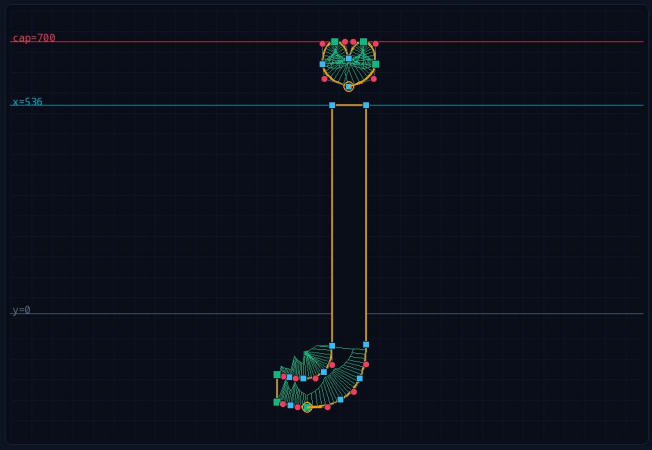
Contour 0 stem x=[85..173], y=[0..536] + Contour 1 heart y=[584..700]



Contours: 2 • On-Curve ■: 10 • Off-Curve ●: 6 • Implied ◆: 0 • AdvW: 258
BBox: x=[61..197] (136 UPM) y=[0..700] (700 UPM)

06 / PHILOCARDIA 'j.heart' (ss07)

Descender hook yMin=-182 UPM + 48 UPM air gap above stem top (y=536)



Contours: 2 • On-Curve ■: 19 • Off-Curve ●: 15 • Implied ◆: 0 • AdvW: 258
BBox: x=[-55..197] (252 UPM) y=[-240..700] (940 UPM)

CURVE STUDY III • ARCHITECTURAL TERMINALS | 25 UPM FELT-MARKER RADII vs SLAB SHELVES | MULTI-CONTOUR INLINE & OUTLINE

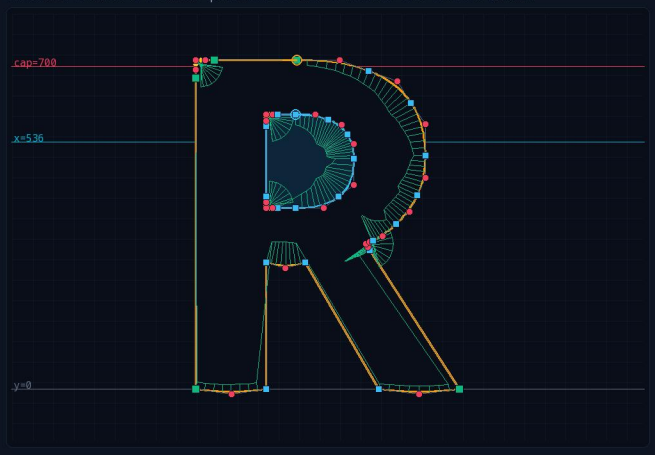
TERMINAL & DISPLAY ARCHITECTURE

Comparing terminal vertex geometry across Soft-Bold (25 UPM radii), Slab-Bold, Chiseltip, MarkerRaw, Outline & Inline

COMPARATIVE DUCTUS: Soft-Bold = 25 UPM 3-node fillets • Slab-Bold = orthogonal serif shelves • Inline = negative inner core channels

01 / SOFT-BOLD 'R' (25 UPM Corner Fillets)

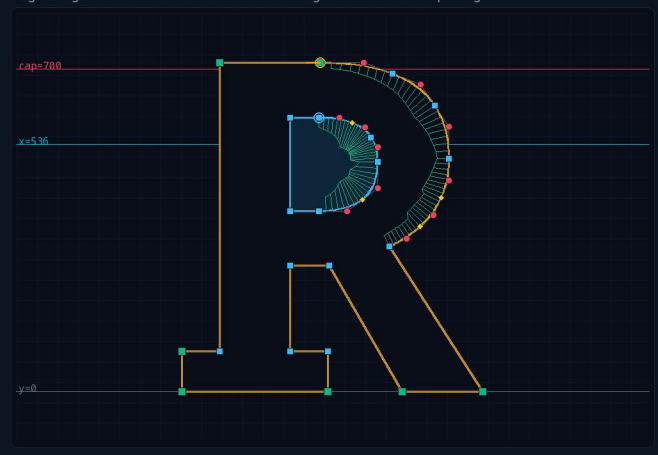
Pillowed counter & rounded quadratic terminal fillets for tactile warmth



Contours: 2 • On-Curve ■: 26 • Off-Curve ●: 26 • Implied ◆: 8 • AdvW: 697
BBox: x=[85..657] (572 UPM) y=[-12..714] (726 UPM)

02 / SLAB-BOLD 'R' (Orthogonal Slab Shelves)

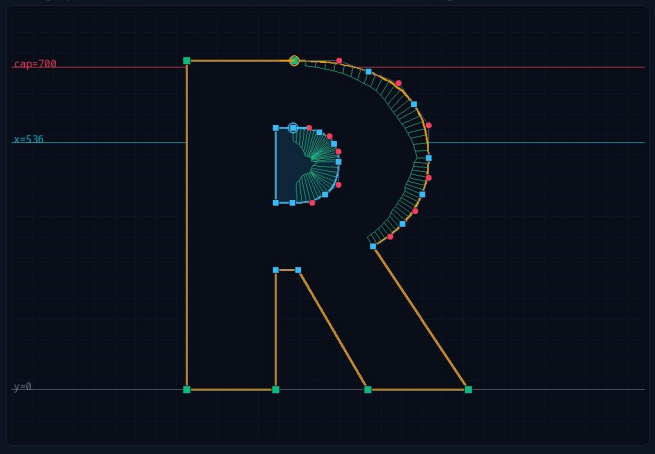
High-weight horizontal slab feet anchoring baseline and cap-height



Contours: 2 • On-Curve ■: 22 • Off-Curve ●: 11 • Implied ◆: 4 • AdvW: 734
BBox: x=[40..694] (654 UPM) y=[0..714] (714 UPM)

03 / CHISELTIP 'R' (45° Broad-Edge Nib)

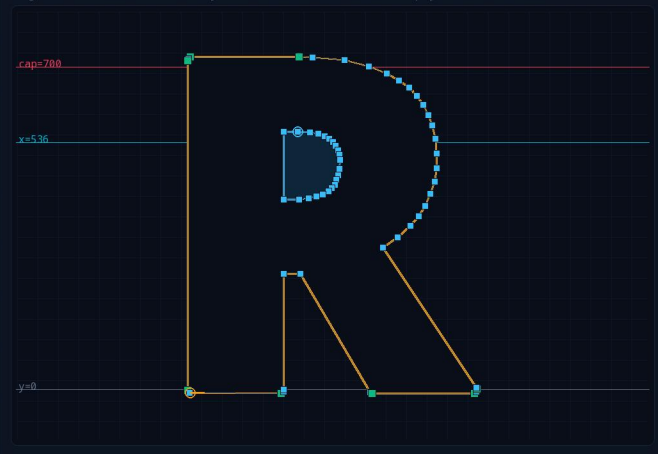
Calligraphic stroke contrast & ribbon terminal modulation (wght 900)



Contours: 2 • On-Curve ■: 22 • Off-Curve ●: 11 • Implied ◆: 0 • AdvW: 729
BBox: x=[77..689] (612 UPM) y=[0..714] (714 UPM)

04 / MARKER-RAW 'R' (Hand-Drawn Cardstock)

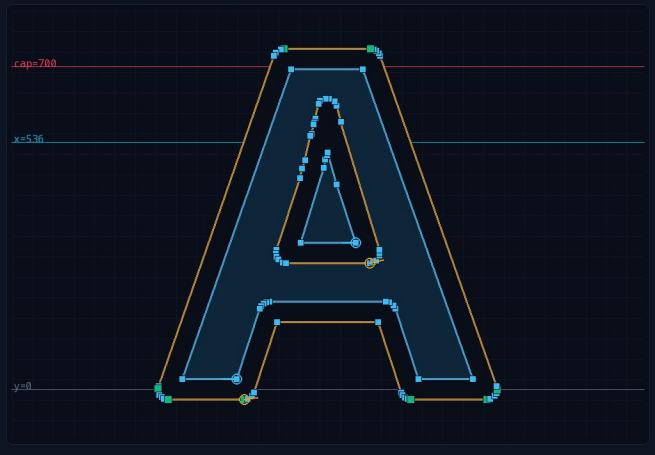
Organic felt-marker velocity & elevated waist ratio on physical cardstock



Contours: 2 • On-Curve ■: 60 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 736
BBox: x=[73..701] (628 UPM) y=[-8..722] (730 UPM)

05 / OUTLINE 'A' (Hollow Dual-Wall Contour)

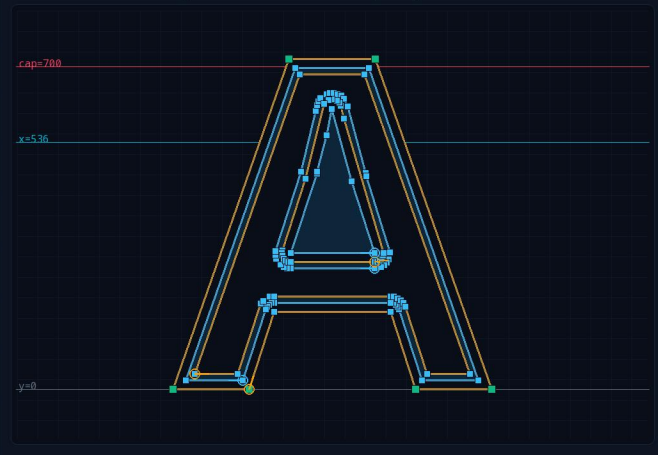
Outer CW boundary paired with parallel interior CCW hollow channel



Contours: 4 • On-Curve ■: 87 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 732
BBox: x=[0..736] (736 UPM) y=[-22..739] (761 UPM)

06 / INLINE 'A' (Chiseled Core Striation)

Multi-contour inline negative core relief inside primary structural stems



Contours: 6 • On-Curve ■: 95 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 732
BBox: x=[0..692] (692 UPM) y=[0..717] (717 UPM)

CURVE STUDY IV • SOVEREIGN INDIGENOUS SCRIPTS | UNDRIP ARTICLES 11, 13, 14, 24 & 31 | 100% SEPARATED FROM TACTILE BRAILLE

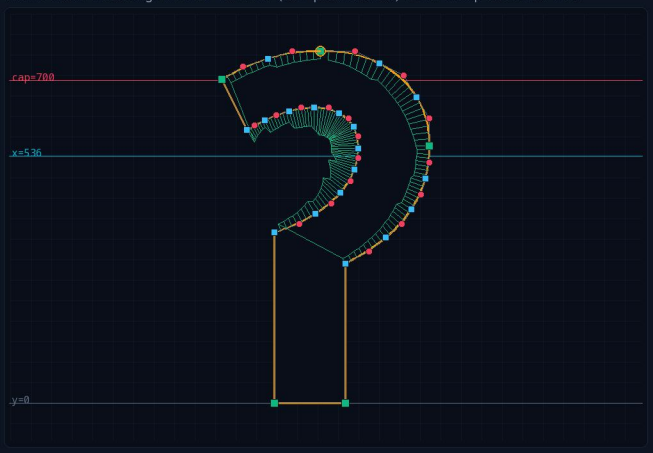
SOVEREIGN INDIGENOUS CURVE TOPOLOGY

Direct vector inspection of Pan-Tribal orthographies (ʔ, ƛ, ŋ, ǫ), Canadian Syllabics (∇) & Cherokee Syllabary (Ꮳ)

SOVEREIGN VECTOR PARITY: Glottal Stop ʔ (U+0294) • Barred Lambda ƛ (U+019B) • Eng ŋ (U+014B) • O-Ogonek ǫ (U+01EB) • Syllabics E ∇ (U+1401) • Cherokee Tsa Ꮳ (U+13E3)

01 / GLOTTAL STOP 'ʔ' (U+0294 • Pan-Tribal)

First-class sovereign letter contour (not punctuation) with crisp baseline stem



Contours: 1 • On-Curve ■: 23 • Off-Curve ●: 19 • Implied ◆: 0 • AdvW: 495
BBox: x=[4..455] (451 UPM) y=[0..765] (765 UPM)

02 / BARRED LAMBDA 'ƛ' (U+019B • Coast Salish)

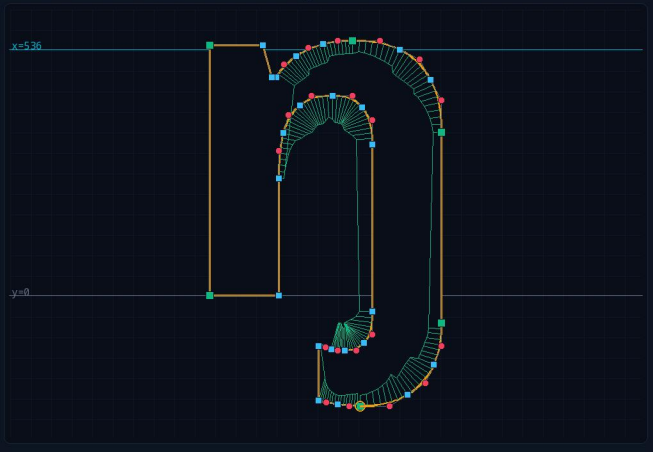
High-contrast crossbar (>= 1.4x aperture) across diagonal descending stroke



Contours: 1 • On-Curve ■: 40 • Off-Curve ●: 25 • Implied ◆: 0 • AdvW: 650
BBox: x=[4..610] (606 UPM) y=[-10..766] (776 UPM)

03 / VELAR NASAL ENG 'ŋ' (U+014B • Lakota/Dene)

Continuous arch transition into descending retroflex/velar J-hook



Contours: 1 • On-Curve ■: 29 • Off-Curve ●: 20 • Implied ◆: 0 • AdvW: 650
BBox: x=[73..579] (506 UPM) y=[-240..556] (796 UPM)

04 / NASAL O-OGONEK 'ǫ' (U+01EB • Diné/Apache)

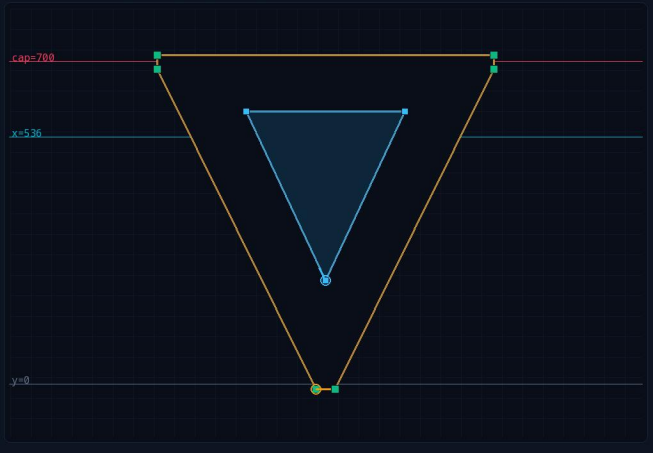
Composite/contour harmony: rightward ogonek hook anchored to lower bowl



Contours: 3 (Composite) • On-Curve ■: 40 • Off-Curve ●: 38 • Implied ◆: 0 • AdvW: 625
BBox: x=[47..577] (530 UPM) y=[-232..556] (788 UPM)

05 / SYLLABICS E '∇' (U+1401 • Inuktitut/Cree)

Rotational triangular apex geometry with balanced interior counter



Contours: 2 • On-Curve ■: 9 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 817
BBox: x=[43..774] (731 UPM) y=[-12..714] (726 UPM)

06 / CHEROKEE TSA 'Ꮳ' (U+13E3 • Tsalagi)

Sequoyah syllabary calligraphic curves rendered in pure quadratic B-splines



Contours: 1 • On-Curve ■: 35 • Off-Curve ●: 30 • Implied ◆: 0 • AdvW: 739
BBox: x=[55..704] (649 UPM) y=[-10..724] (734 UPM)

CURVE STUDY V • ISO 17049 TACTILE BRAILLE | r = 60 UPM QUADRATIC DOME CIRCLES | GROUNDED ON BASELINE y = 0 UPM

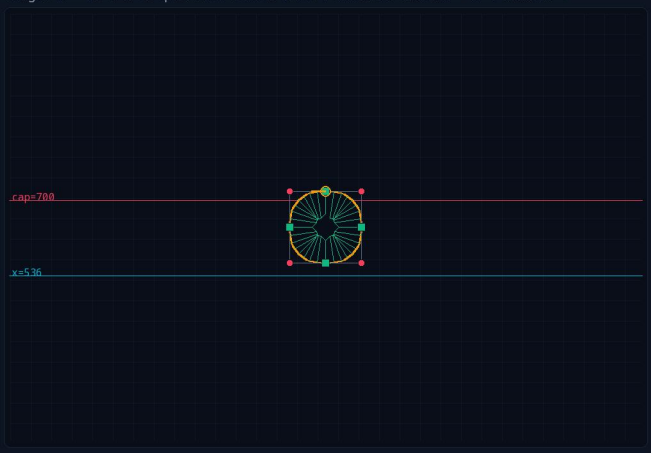
TACTILE BRAILLE & SCIENTIFIC TOPOLOGY

Dedicated inspection of ISO 17049 6-dot/8-dot circular dome geometry (U+2800–U+28FF) & multi-contour scientific symbols

TACTILE & SCIENTIFIC PRECISION: Dot-1 ⠁ (U+2801) • Dots-123 ⠇ (U+2807) • 6-Dot Full ⠿ (U+283F) • 8-Dot Full ⣿ (U+28FF) • Music Sharp ♯ • Copyright ©

01 / BRAILLE DOT-1 '⠁' (U+2801 • Letter A / 1)

Single r = 60 UPM CW quadratic circle with 4 cardinal extrema ■ & 4 handles ●



Contours: 1 • On-Curve ■: 4 • Off-Curve ●: 4 • Implied ◆: 0 • AdvW: 750
BBox: x=[132..288] (156 UPM) y=[564..720] (156 UPM)

02 / BRAILLE DOTS-123 '⠇' (U+2807 • Letter L)

Vertical column alignment: uniform 2.5 mm intra-cell dot pitch



Contours: 3 • On-Curve ■: 12 • Off-Curve ●: 12 • Implied ◆: 0 • AdvW: 750
BBox: x=[132..288] (156 UPM) y=[0..720] (720 UPM)

03 / 6-DOT LITERARY CELL '⠿' (U+283F • Dots 1-6)

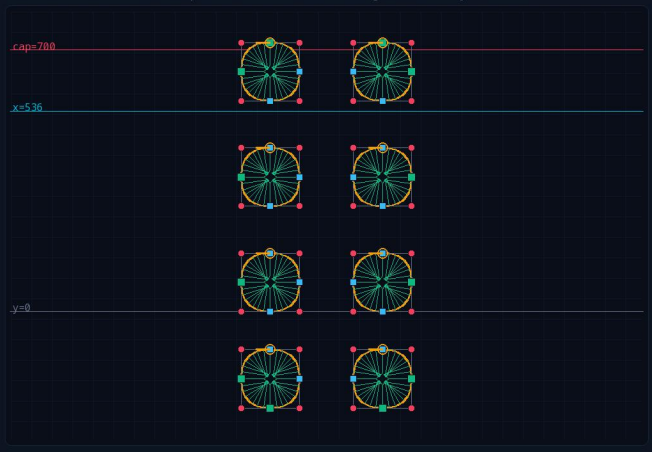
Standard 2x3 literary/pharmaceutical embossing grid (6 independent CW contours)



Contours: 6 • On-Curve ■: 24 • Off-Curve ●: 24 • Implied ◆: 0 • AdvW: 750
BBox: x=[132..588] (456 UPM) y=[0..720] (720 UPM)

04 / 8-DOT TELEMETRY CELL '⣿' (U+28FF • Dots 1-8)

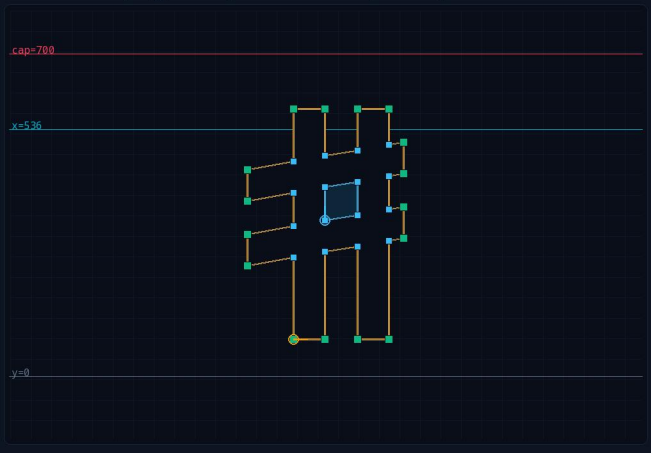
Full 2x4 ISO/TR 11548 computer & ICU Braille cell grounded at y = 0 UPM



Contours: 8 • On-Curve ■: 32 • Off-Curve ●: 32 • Implied ◆: 0 • AdvW: 750
BBox: x=[132..588] (456 UPM) y=[-258..720] (978 UPM)

05 / MUSIC SHARP '♯' (U+266F • PocketGull-Music)

Multi-contour topology: outer grid + interior CCW parallelogram counters



Contours: 2 • On-Curve ■: 32 • Off-Curve ●: 0 • Implied ◆: 0 • AdvW: 540
BBox: x=[120..460] (340 UPM) y=[80..580] (500 UPM)

06 / COPYRIGHT '©' (3-Tier Nested Winding)

Outer ring (CW) -> Inner ring (CCW) -> Central letter C (CW) parity



Contours: 3 • On-Curve ■: 51 • Off-Curve ●: 49 • Implied ◆: 0 • AdvW: 832
BBox: x=[49..783] (734 UPM) y=[-10..724] (734 UPM)